

1. What is the primary function of an ALU in a digital system?

The primary function of an Arithmetic Logic Unit is to perform all arithmetic operations and logical operations on binary data.

2. Explain the difference between arithmetic and logical operations in the context of an ALU.

Arithmetic operations involve mathematical calculations on numbers. While logical operations involve bitwise manipulation of data, where each bit is processed based on logical rules rather than numeric value.

3. Why is it necessary to have a control signal (like Operation in your lab) for an ALU?

The control signal tells the ALU which specific operation to perform at a given time. Since the ALU can perform many different operations, the control signal selects one of them. Without this signal, the ALU would not know whether it should add, subtract, AND, OR, etc.

4. What is the purpose of a testbench in Verilog?

A testbench is used to simulate and verify the behavior of your Verilog design. It provides input signals, runs the module, observes outputs, and helps ensure that your design works correctly before it is synthesized into hardware.

5. Describe the role of the decoder module in your ALU design.

The decoder translates a binary operation code into individual control lines that enable the correct operation inside the ALU. For example, if the Operation input is 00, the decoder activates the circuit for AND, while if it is 01, it activates OR.

Waveform

